6(a) Binding energy is defined as the amount of energy needed to split a nucleus into its individual nucleons.

(b) **(i)** ${}^{4}_{2}He \rightarrow {}^{3}_{2}He + {}^{1}_{0}n$

(ii) Q = Difference in total BE [CONCEPT]

= 4(6.8465) – 3(2.2666)

= 20.5862 MeV

(iii) mass of neutron + mass of ${}^{3}_{2}He$ – mass of ${}^{4}_{2}He$ = $\frac{Q}{931.494}$

mass of ${}^{4}_{2}He$ – mass of ${}^{3}_{2}He$ = mass of neutron - $\frac{Q}{931.494}$

= 1.0097 u - $\frac{20.5862}{931.494}$

= 0.9876 u

(iv) Since energy is supplied to make the process happen, this is a mass excess reaction.
So $m_{helium\text{-}4} < m_{helium\text{-}3} + m_{neutron}$
$m_{helium\text{-}4} - m_{helium\text{-}3} < m_{neutron}$

7 (a) Newton's law of gravitation states that the force of attraction between two point masses is